

Introduction to TCP Congestion Control

TCP congestion control prevents network overload by limiting the amount of data a sender can transmit before receiving an acknowledgment. The core mechanisms are:

1. Slow Start: The congestion window (cwnd) begins at 1 MSS and doubles each RTT until a threshold or packet loss is detected.
2. Congestion Avoidance: Once cwnd reaches ssthresh, it increases linearly (additive increase) rather than exponentially.
3. Fast Retransmit: If three duplicate ACKs arrive, the sender retransmits the lost segment without waiting for a timeout.
4. Fast Recovery: After fast retransmit, the sender enters fast recovery instead of slow start, keeping the connection throughput higher.

The Tahoe and Reno variants differ in how they handle packet loss: Tahoe always drops to slow start after loss, while Reno uses fast recovery to avoid full restarts.